

15	B	Using Malus' law, the intensity of transmitted light (I) through polarisers is proportional to $I_0 \cos^2 \theta$, where I_0 is the intensity of transmitted through the first polariser, θ is the angle between two polarisers. Unpolarized light loses exactly half of its intensity after it passes through polariser W no matter what the direction of polarising for polariser X is: $0.50 I_0$. The angle between polariser W and X is 10°, the transmitted intensity after X would be $0.50 I_0 \times \cos^2 (40^\circ - 30^\circ) = 0.48 I_0$. The angle between polariser X and Y is 20°, the transmitted intensity after Y would be $0.48 I_0 \times \cos^2 (60^\circ - 40^\circ) = 0.43 I_0$.
16	B	Since the wave is travelling from left to right, at the next instant, every particle will "copy"